# The Human Reduction Problem

As we approach an era where the societies around us manifest themselves as the societies of control depicted by Deleuze, it becomes clearer that these societies reduce its members to predictive functions or data mines. These systems are designed around the idea that we can model humans not necessarily as autonomous individuals but rather a being defined by their past actions and their predictable future ones. This paper will analyze the systemic reductive process of reducing humans to mere components in a larger technical system, this process creates a rather successful system at the expense of dehumanizing the humans involved in the process.

First, what is reductionism? According to Alisa Ney's entry in the Internet Encyclopedia of Philosophy, "reductionists are those who take one theory or phenomenon to be reducible to some other theory or phenomenon....a reductionist about biological entities like cells might take such entities to be reducible to collections of physico-chemical entities like atoms and molecules"[1]. For the context of this paper, when saying a human is reducible to a simple component in a larger system, this means that the human behaviour in the context of the system can be explained as a single task or component of the system. In otherwords, human behaviour/actions/outcomes are explainable by a set of known rules and facts. This is slightly different then a lot of the more formal definitions of redicibility where Theory A is reducible to Theory B if everything in Theory A is derivable from Theory B, but the overall idea remains the same. Now there are some critics when it comes to reductionism as a whole, with Richard Jones noting that "the view that we should cut up reality to get to its 'real' components has been a part of Western philosophy since its beginnings. But so has the competing more integrative approach of antireductionism"[2]. According to Jones, antireductionism was part of Aristotle's philosophy where "he [Aristotle] also introduced the famous phrase 'The whole is different from the sum of its parts.' Substance-, structure-, theory-, and method-reductionisms all fit well with the atomic approach but are hard to reconcile with the Aristotelian approach"[3]. Here it is evident that according to Jones, Aristotle is strictly against the reductive approach and this Aristotelian holistic approach dominated over the atomistic approach for many centuries[4].

However,Lewis Mumford believes there were instances of the reduction of humans dating back to even before Aristotle and the ancient greeks. Mumford made notions of a historic machine that was a systematic reduction of humans into components. According to Lewis, this machine was never really discovered for a long time "for a simple reason: it was composed almost entirely of human parts. These parts were brought together in a hierarchical organization under the rule of an absolute monarch... Let us call this archetypal machine—the human model for all later specialized machines—the 'Megamachine' "[5]. These "Megamachines" were

[1]Alissa Ney, "Reductionism," *Internet Encyclopedia of Philosophy*.

[2]Richard Jones, *Reductionism: Analysis and the Fullness of Reality* (Bucknell University Press, 2000), 2.

[3]Jones, *Reductionism: Analysis and the Fullness of Reality*, 3.

[4]Jones, *Reductionism: Analysis and the Fullness of Reality*, 3.

[5]Lewis Mumford, "Technics and the Nature of Man," *Technology and Culture* 7, no. 3 (Summer 1966): 312.

complex systems that were used to materialize the will of those who are in charge of them, to create things no single part could create. For these machines, efficiency came at the cost of simplifying the human workers into smaller parts and tools for the machine itself. According to Mumford, "by operating as a mechanical unit, the 100,000 men who worked on that pyramid generated 10,000 horsepower; this human mechanism alone made it possible to raise that colossal structure with the use of only the simplest stone and copper tools"[6]. These machines were how the great pyramids in egypt were made and many other colossal structures. For Mumford, these same machines are used in the contemporary scene with the military and other large corporations. Similarly, Peter Galison described an elaborate process to create antiaircraft predictors. In creating these machines, they started to view the enemy human pilots as a "servo-mechanism" which helped determine the path of the aircraft. Galison quoted Wiener's "Summary Report for Demonstration" where Wiener noted that "the pilot behaves like a servo-mechanism, attempting to overcome the intrinsic lag due to the dynamics of his plane as a physical system"[7]. Here, Wiener choose to view the pilot as a mere part in a system, designed to correct and guide the aircraft, rather than a human with fully autonomous capabilities. Essentially, this project aimed to reduce all of the pilots years of training and experience to a single predictable mechanism solely meant to overcome the lag and error of the system. In fact, Wiener viewed this reduction as necessary because for Weiner, "It does not seem even remotely possible to eliminate the human element as far as it shows itself in enemy behavior. Therefore, in order to obtain as complete a mathematical treatment as possible of the over-all control problem, it is necessary to assimilate the different parts of the system to a single basis, either human or mechanical. Since our understanding of the mechanical elements of gun pointing appeared to us to be far ahead of our psychological understanding, we chose to try to find a mechanical analogue of the gun pointer and the airplane pilot"[8]. This demonstrates how Wiener viewed the reduction of the enemy pilot to a simple mechanical piece seemed necessary to him. To Wiener, viewing the enemy pilots as human made these systems messy and complicated beyong the current understanding of human psychology. As such, the complexities were discarded in a simpler view of humans as a mechanism in a larger machine. This gave rise to an overall black box view of human behaviour where in the context of a single problem, all the unique human experiences that makes an individual human can be seemingly ignored and reduced to a merely predictable input and output system. Thus, Mumford's primary fear of the overall reliance on the "Megamachine" and tools as a whole is that humans, "instead of functioning actively as a tool-using animal, man will become a passive, machine-serving animal whose proper functions, if this process continues unchanged, will either be fed into a machine, or strictly limited and controlled for the benefit of depersonalized collective organizations"[9].

This leads directly into the 20th century notions of the reduction of the human subject. Gilles

[6]Lewis Mumford, "Technics and the Nature of Man," *Technology and Culture* 7, no. 3 (Summer 1966): 312.

[7]Peter Galison, "The Ontology of the Enemy: Norbert Wiener and the Cybernetic Vision,' *Critical Inquiry* 21, no. 1 (Autumn 1994): 236.

[8]Galison, "The Ontology of the Enemy," 241.

[9]Mumford, "Technics and the Nature of Man," 303.

Deleuze in his "Postscript on the Societies of Control" described "Societies of Control" as an extremely modular system where individuals are assigned passwords or ids which grant or reject access to places, services, or information. This differs from the disciplinary societies which has "the signature that designates the individual, and the number or administrative numeration that indicates his or her position within a mass… In the societies of control, on the other hand, what is important is no longer either a signature or a number, but a code…[which] mark access to information or reject it"[10]. According to Deleuze, these societies of control determine access through viewing individuals as a "dividual", looking at a human into smaller parts of data to be interpreted. For Deleuze, it was scary to no longer look at "the mass/individual pair. Individuals have become 'dividuals,' and masses, samples, data, markets, or 'banks'"[11]. In other words, for the societies of control, people become valuable products for their vast amounts of information. Similarly, Elizabeth Hirschman did a study of how "personal advertisements represent the offering of people as products, as a set of marketable resources in search of an appropriate buyer"[12]. In her study, she primarily focused on a sort of dating scene where people were essentially marketing themselves looking for a prospective romantic partner. This sort of marking perspective created this idea that "all consumer/producer transactions are instrumental in form and intent; that they are based only on calculated, analytic motivations; that we see ourselves and others only as sets of tradable resources"[13]. Tying back to Deleuze, this perspective causes one to look at themselves and see which parts of themselves are valuable or tradable, separating the individual and becomming a dividual. Deleuze hypothesized that the use of dividuals and a data driven society of control will lead to a variety of dystopian services, such as "medicine 'without doctor or patient'", "electronic collars that force the convicted person to stay at home", and even "the introduction of the 'corporation' at all levels of schooling"[14]. The scariest part of Deleuze's Postscript is that some of these dystopian services are being used and implemented today around the world.

In fact, James Brusseau does a deep dive on how Deleuze's concepts are being materialized in the world around us today. Brusseau opens his paper with the idea that "today, 'control' is no longer an abstract concept in political philosophy; it is localisable as specific technologies functioning where personal information is gathered into contemporary data commerce"[15]. More specifically, Brusseau argues that control is formed through predictive analysis in our modern world, where "data-driven marketing and social media strategies that regulate [or control] through incentives"[16]. Essentially, we are in a society that is centered around the use and collection of data and we are rewarded and controlled based on the data we create and the data we forfeit to corporations. In fact, Angela Dennis comments on this very same dillema

[10]Gilles Deleuze, "Postscript on the Societies of Control," *October* 53 (Summer 1990): 5.

[11]Deleuze, "Postscript on the Societies of Control," 5.

[12]Elizabeth Hirschman, "People as Products: Analysis of a Complex Marketing Exchange," *Journal of Marketing* (January 1987): 101.

[13]Hirschman, "People as Products," 106–107.

[14]Deleuze, "Postscript on the Societies of Control," 7.

[15]James Brusseau, "Deleuze's Postscript on the Societies of Control: Updated for Big Data and Predictive Analytics," (2020): 2.

[16]Brusseau, "Updated Deleuze's Postscript on the Societies of Control" 2

where “Deleuze suggests that humans have moved from a discipline-through-laws society in which a human was seen as an individual (as contrasted with a mass), towards a society of continuous control and surveillance in which the human has become ‘dividual’ - parsed into various informational units”[17]. This constant surveillance and modulation of the human as data is quite the scary thing to come to terms with. This sort of surveillance and data can even help the police find criminals who are masters of disguise and evasion. Dennis writes about a killer D’Angelo where “After three decades of evading capture, eluding brilliant investigators, D’Angelo was caught by the brute power of mass data and algorithms, and he never saw it coming”[18]. Specifically, the police had some of D’Angelo’s DNA from a crime scene 3 decades ago. They used this DNA on a genealogical database called GEDMatch, a service designed to help users create a more complete family tree. Essentially, the police used this service to find some of D’Angelo’s potential distant relatives, and along with along with their current known information about the suspect, then narrowing down the suspect to D’Angelo. From the vast amounts of data the police had access to they were able to find and reprimand the suspect. Similarly, there is research being done on predictive genetic testing in which subjects are screened based on their genetic makeup. In other words, if a person is potentially susceptible to a disease then they would get screened for it. However the researchers note that “the effectiveness of interventions targeted at individuals with a specific genetic makeup will be difficult to prove in the short term”[19]. While it is good that the police were able to find the suspect and predictive genetic testing may save lives, these cases have much darker implications for our society as a whole. Dennis notes this very same thing in her article, “D’Angelo’s capture offers some solace to his living victims and the families of the deceased. But this case has wider implications – it reveals the power of the web of ever-increasing digitised data about each of us, which may be classified, parsed up, separated, and recombined ad infinitum, and which is able to be used, and is being used, in a myriad of unknown and unknowable ways, both for and against us”[20]. Stephan Scipal notes that in this new data-driven age, “We are not just witnessing technological change; we are experiencing the emergence of a new kind of megamachine, one that promises efficiency and progress while fundamentally altering what it means to work as a human being”[21]. Scipal compares the current modern autommized workforce to Mumford’s Megamachine, where the humans within the system are reduced to mere componentes in the name of efficiency. Scipal notes that “The Al-powered workplace operates on strikingly similar principles [to Mumford’s Megamachine], but with algorithmic precision that would have astounded even the most ambitious industrial engineers of the past”[22]. With the rise of all sorts of new technologies, the ways we can be controlled and our very definitions of what it means to be an efficient worker are drastically changing.

[17]Angela Dennis, “The Golden State Killer & Deleuze’s ‘Dividual’,” *Philosophy Now* (2022): 2.

[18]Dennis, “The Golden State Killer”, 2

[19]Steve Morgan, Jeremiah Hurley, Fiona Miller, and Mita Giacomini, “Predictive genetic tests and health system costs,” *Canadian Medical Association Journal* (2003): 1.

[20]Dennis, “The Golden State Killer”, 2

[21]Stephan Scipal, “The New Megamachine: How Augmented Intelligence is Reshaping Our Working Lives,” *LinkedIn*, last modified July 25, 2025, 2.

[22]Scipal, “The New Megamachine”, 2

In conclusion, the trajectory from the ancient Megamachine to the algorithmic societies of control reveals a persistent and troubling legacy of human reduction. This process, which began by simplifying human labor for colossal construction, has now evolved into a pervasive data-driven paradigm that parses individuals into divisible, marketable units. The warnings of Mumford, Wiener, and Deleuze have materialized in a world where predictive analytics and genetic surveillance increasingly define and confine human potential. As we stand at this crossroads, the central challenge is no longer merely to critique the system but to actively decide if the price of efficiency, the erosion of human autonomy and complexity, is one we are willing to continue paying. The future of our society depends on whether we can reclaim the individual from the dividual and resist the seductive, yet dehumanizing, logic of the megamachine in its latest, most powerful form.

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